2024 IEEE 5th International Conference on Technology, Informatics, Management, Engineering & Environment (TIME-E)
Bali, Indonesia, August 7-9, 2024

Potato Leaf Disease Detection by Means of Vision Transformer and CoAtNet Architecture

Mesa Anugerah Kenan Sergio
Engineering Physics Study Program
Institut Teknologi Bandung
Bandung, Indonesia
13320057@mahasiswa.itb.ac.id

Ariel Brandon Stevano Siallagan
Engineering Physics Study Program
Institut Teknologi Bandung
Bandung, Indonesia
13320016@mahasiswa.itb.ac.id

Endra Joelianto
Instrumentation, Control, and Automation Research Group
Institut Teknologi Bandung
Bandung, Indonesia
ejoel@itb.ac.id

2024 IEEE International Conference on Technology, Informatics, Management, Engineering and Environment (TIME-E) | 979-8-3503-6475-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/TIME-E62724.2024.10919496

***Abstract*—Potatoes are a food source that is beneficial for humans in various aspects, such as having nutritional and economic value. However, potato plants are susceptible to many diseases. To address this issue, various researchers have suggested methods utilizing machine learning and image processing for automatic disease type detection. This paper introduces a disease detection model for potato leaf images, employing two computer vision techniques: Vision Transformer (ViT) and CoAtNet. The dataset used in this paper consisted of 30,760 augmented images divided into seven classes. The ViT model had a test accuracy of 99.32% and a validation accuracy of 99.12%. Meanwhile, the CoAtNEt model had a test accuracy of 99.58% and a validation accuracy of 99.48%. These results show that both models can be used in the detection of potato leaf diseases.**

Keywords—potato, leaf disease, image processing, vision transformer, CoAtNet

I. Introduction

Agriculture is the foundation of global food security and economic stability, with its growth and development being crucial in supporting the continuously growing human population [1]. Technological advancements in this sector have significantly increased crop yields and quality, helping to meet the rising food demand. Among various crops, potatoes hold a vital position due to their great impact on the economic and nutritional status of people worldwide [2]. Potatoes serve as a staple food for millions of people, providing essential vitamins, minerals, and energy.

Potato cultivation is crucial for agricultural productivity, but it faces numerous challenges, especially in disease management. Potato plants are highly susceptible to various diseases, which significantly impact their yield and quality. Among these diseases, those affecting the leaves are particularly detrimental because they directly impair the plant's photosynthetic capacity, thereby hindering tuber growth and development. Effective and timely detection of potato leaf diseases is essential to mitigate these impacts and ensure healthy crop production [3].

Diseases affecting potato leaves can be caused by bacteria (*Ralstonia solanacearum*), fungi (*Phytophthora infestans*), nematodes, pests, and viruses. Each type of pathogen exhibits unique symptoms that can lead to yield losses if not detected and managed promptly [4]. Two examples are common fungal diseases such as late blight (caused by *Phytophthora infestans*) and early blight (caused by *Alternaria solani*). Late blight is notorious for its rapid spread and devastating impact on crop yields. Early blight, although less aggressive, can still significantly reduce yields by causing lesions on leaves and stems [5]. Therefore, developing techniques for the early detection of potato leaf disease is crucial for enhancing productivity and supporting sustainability potato farming.

Previous research has explored various algorithms and technologies for detecting potato leaf diseases. Conventional methods, such as visual inspection and laboratory tests, have been the cornerstone, but are time-consuming and require trained personnel to carry out. Recent advancements leverage artificial intelligence (AI) and computer vision to automate disease detection processes [6]. These technologies promise more efficient disease identification and have the potential to revolutionize agricultural practices, especially in potato cultivation.

In this study, a model was designed to detect potato leaf diseases based on images using two advanced machine learning models, namely Vision Transformer (ViT) and CoAtNet. ViT is a model that relies on a transformer architecture [7] that has proven to be very effective in various image processing tasks. ViT is able to utilize self-attention capabilities to capture important features from potato leaf images, thereby providing excellent classification performance. On the other hand, CoAtNet combines convolutional neural networks (CNNs) with attention mechanisms to enhance efficiency and accuracy in image recognition tasks [8]. By combining the advantages of CNN in local feature extraction and the advantages of attention mechanisms in capturing global relationships, CoAtNet can produce superior performance in detecting various types of diseases on potato leaves. These two models were successfully tested and compared based on the dataset from [9], which contains images of potato leaves infected with various diseases. The dataset contains seven classes, which distinguishes it from other datasets.

This success emphasizes the great potential of applying machine learning technology in the agricultural sector, especially for accurately detecting plant diseases automatically. Implementation of these models can help farmers and agronomists monitor plant health more efficiently, so that preventive measures can be taken earlier and agricultural productivity can be increased. This research also paves the way for further development of similar models that can be adapted to a variety of conditions and other types of crops, which could ultimately contribute to improving global food security.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:10 UTC from IEEE Xplore. Restrictions apply.

II. Related Work

Several studies on the identification of disease on potato leaves have been done previously. In [10], Sholihati et al. used VGG19 and VGG16 models for detecting potato leaf diseases. The experiment achieved a test accuracy of 91.31% and 90.96% for VGG 16 and VGG19, respectively. The study by Irmawati et al. [11] used EfficientNet and the ConvNeXt architecture on a dataset that was divided into five classes. The test accuracy was 95.92% and 94.29% for EfficientNet and ConvNeXt, respectively. Islam et al. [12] used image segmentation and the multiclass support vector machine method to detect potato leaf diseases. This method had an accuracy of 95%. Hylmi et al. [13] utilized image segmentation and Multi-Class Support Vector Machine. They extracted features from segmented images, including RGB color histograms, texture features using the Gray Level Co-occurrence Matrix (GLCM) and shape features by calculating leaf spot contours. This system achieved an accuracy of 97.56%. Wasalwar et al. [14] in 2023 also used CNN with a dataset consisting of 3,000 samples. The CNN-based model had an accuracy of 99%. Srivastava et al. [15] used a YOLOv7 model for detection and classification of potato leaves. Bajpai et al. [16] used three deep learning models, namely VGGNet16, ResNet101, and AlexNet, with the modified AlexNet having the best accuracy, at 99.97%. In [17], Hoysala et al. used CNN and a dataset consisting of 1,000 images to create a model. The model had 99.07% testing accuracy. In [18], Bangari et al. used Artificial Neural Network (ANN), Support Vector Machine (SVM), and Convolutional Neural Network (CNN) to detect potato leaf diseases. The model was trained and validated using 2,150 images resulting in 85% accuracy for ANN, 88.89% accuracy for SVM, and 99.07% accuracy for CNN.

III. Methodology

This research proposes a strategy for detecting disease in plant leaf images, as shown in Fig. 1. The research started with data collection and pre-processing. Next, the pre-processed data was divided into three sets: for training, validation, and testing, respectively. After training and evaluating the model with these datasets, the next step involved assessing its performance using the training and validation datasets. In the final stage, the predictive model was tested with the testing dataset to evaluate how well it was trained.

A. Data Collection and Pre-processing

This research used the dataset from [9], which consists of seven classes with a total of 3,076 images, depicted in Fig. 2. The classes were categorized as “Bacteria”, “Fungi”, “Healthy”, “Nematode”, “Pest”, “Phytophthora”, and “Virus”. Images in this dataset were taken in unrestricted settings, like different backgrounds and varying distance. To improve the number of images in the dataset, each image in the dataset was augmented ten times using the imgaug library in Python. After augmentation, the size of the dataset increased into 30,760 images.

B. Split Dataset

The dataset was divided into subsets at a ratio of 80:10:10 for training, validation, and testing. Table I lists the data for training, validation, and testing for each class.

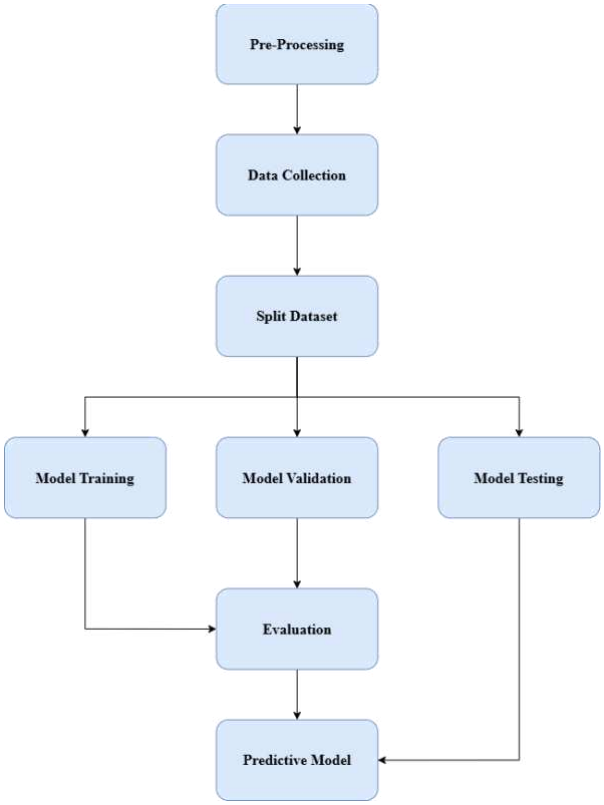

Fig. 1. Block diagram of the proposed methodology.

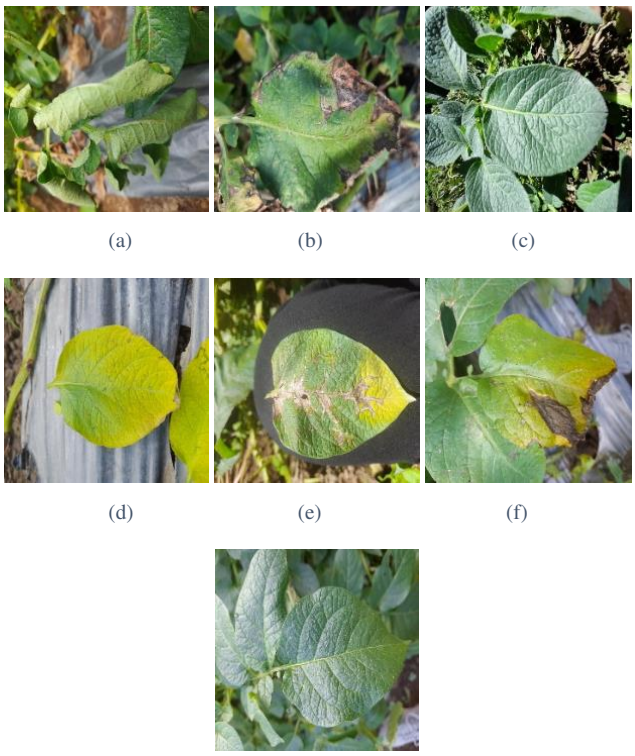

Fig. 2. Samples from the dataset: (a) bacteria; (b) fungi; (c) healthy leaf; (d) nematode; (e) pest; (f) phytophthora, (g) virus.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I. DATASET FOR TRAINING, VALIDATION, AND TESTING AFTER AUGMENTATION

Class	Dataset		
	Training	*Validation*	*Testing*
Bacteria	4552	569	569
Fungi	5984	748	748
Healthy	1608	201	201
Nematode	544	68	68
Pest	4888	611	611
Pythophthora	2776	347	347
Virus	4256	532	532
Total	24608	3076	3076

C. *Model Training and Testing*

For this research, the models that were used were Vision Transformer (ViT) and CoAtNet. The models were trained using the transfer learning method.

The ViT architecture was introduced in [7]. ViT was the first architecture to rely solely on transformers for image classification, removed the necessity for any convolutional layers. The architecture delivered state-of-the-art performance on numerous benchmarks [19]. The architecture of ViT is presented in Fig. 3.

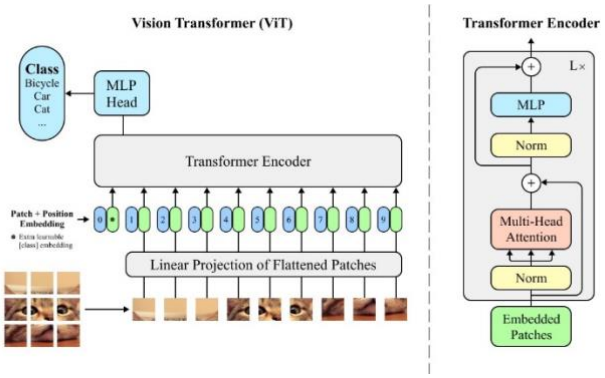

Fig. 3. Architecture of ViT.

The CoAtNet architecture, represented in Fig. 4., was introduced in [8]. A CoAtNet model is a combination of depthwise convolution and self-attention layers to improve generalization, capacity, and efficiency. Principally, CoAtNet models combine a ViT architecture with a CNN architecture. A CoAtNet model has two main components: a mobile inverted bottle-neck convolution (MBConv) block [20] and self-attention with relative bias [21].

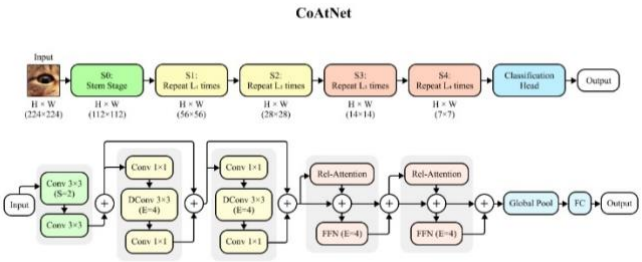

Fig. 4. Architecture of CoAtNet.

In this research, we used a ViT model from Google, which is available on Hugging Face [22]. The ViT model was trained on ImageNet-21k at a resolution of 224 x 224 pixels. Using ViTFeatureExtractor and ViTForImageClassification from the Transformers library, we extracted features from the model. The CoAtNet model, also available on Hugging Face [23], was trained in timm on ImageNet-1k by Ross Wightman at a resolution of 224 x 224 pixels. Using the timm library, we could create and extract the model. The batch size for both models was 32. For training, three epochs were used with one optimizer, namely Adam.

The performance of these models in detecting several types of leaf diseases was evaluated using several metrics, i.e., accuracy, precision, recall, and F1-score. The equation for each metric is shown in the following equations:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

$$F1-score = \frac{2\times Precision\times Recall}{Precision+Recall} \quad (4)$$

"The best model trained on the dataset was tested with the testing dataset to measure its performance. In this research, we tested the model using 3,076 images from the testing dataset.

IV. RESULT AND ANALYSIS

A. *Model Training and Validation*

For training, three epochs were used to prevent overfitting. After training, we got the training and validation loss for each model (see Fig. 5. and Fig. 6.). From these figures, it can be seen that the training loss was lower than the validation loss, although the difference was minimal. Additionally, both models exhibited a stable validation loss, suggesting that overfitting was not very significant.

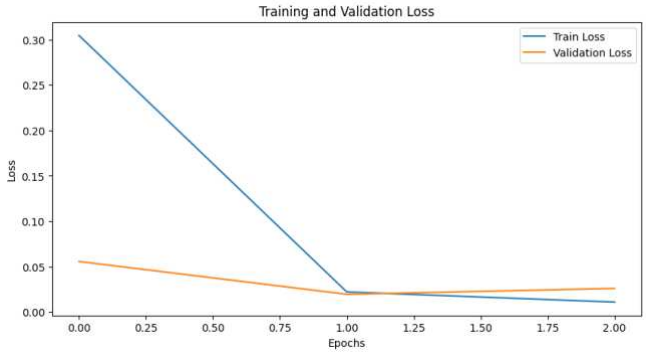

Fig. 5. Training and validation loss for the ViT model.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:10 UTC from IEEE Xplore. Restrictions apply.

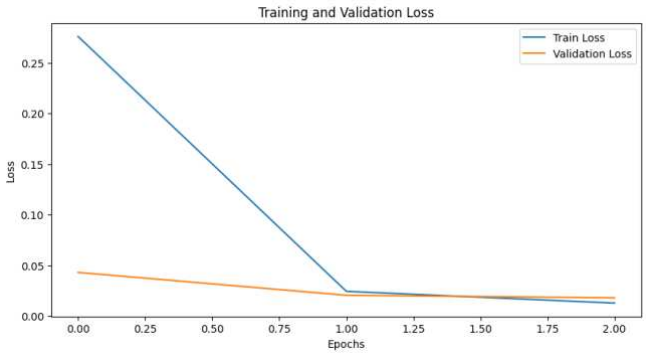

Fig. 6. Training and validation loss for the CoAtNet model.

Figures 7 and 8 show graphs of the training and validation accuracy values for each model.

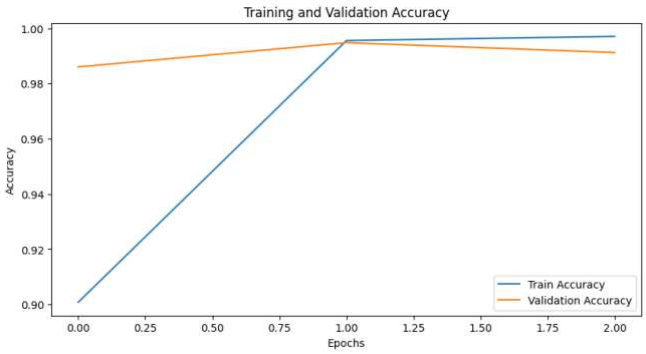

Fig. 7. Training and validation accuracy for the ViT model.

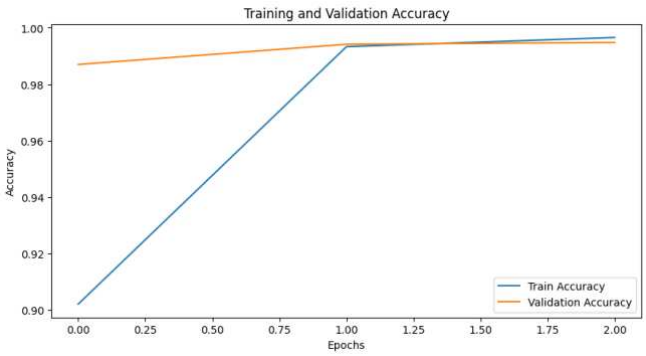

Fig. 8. Training and validation accuracy for the CoAtNet model.

TABLE II. VIT MODEL CLASSIFICATION REPORT

Class	Classification Report		
	Precision	***Recall***	***F1-score***
Bacteria	1.00	1.00	1.00
Fungi	1.00	0.98	0.99
Healthy	1.00	0.99	0.99
Nematode	1.00	0.97	0.99
Pest	0.97	1.00	0.98
Pythophthora	1.00	1.00	1.00
Virus	1.00	1.00	1.00

B. Evaluation

One way to evaluate a model's performance is by using a classification report. A classification report shows several metrics, i.e., precision, recall, and F1-score, for each class. A classification report for each model is presented in Tables II and III. They were visualized using a spider chart, as shown in Fig. 7. and Fig. 8., respectively. Based on the tables and figures below, both models had excellent precision, recall, and F1-scores.

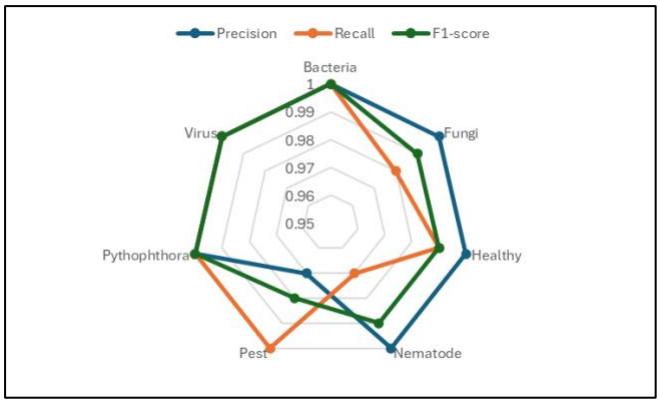

Fig. 9. Visualization of the ViT model classification report.

TABLE III. COATNET MODEL CLASSIFICATION REPORT

Class	*Classification Report*		
	Precision	***Recall***	***Precision***
Bacteria	1.00	1.00	1.00
Fungi	0.99	1.00	1.00
Healthy	1.00	0.98	0.99
Nematode	1.00	0.97	0.99
Pest	0.99	0.99	0.99
Pythophthora	1.00	1.00	1.00
Virus	1.00	1.00	1.00

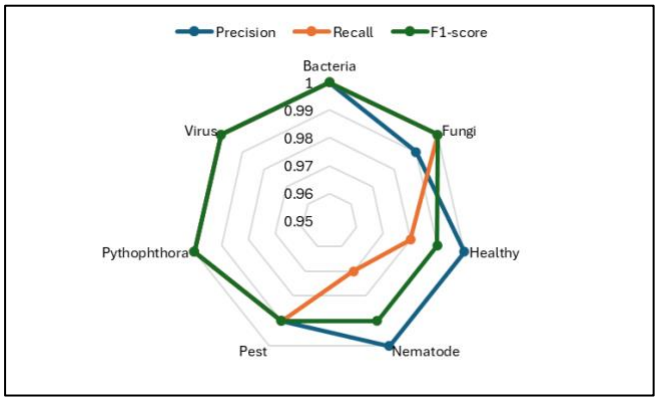

Fig. 10. Visualization of the CoAtNet model classification report.

From the tables and figures above, the models achieved excellent precision, recall, and F1-score for all classes, especially on the bacteria and virus classes. High precision means the models made few false positive errors, high recall means the models made few false-negative errors, and a high F1-score means the models had balance in precision and recall.

Another way of evaluating the results is by using a confusion matrix. A confusion matrix compares the predicted and the actual values of a dataset. Here, the confusion matrix used a test dataset consisting of 3,076 images, which could provide details of how the model performed. Figures 9 and 11 show the confusion matrixes for both models. Examples of predictions from the models are shown in Figures 10 and 12, respectively.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:10 UTC from IEEE Xplore. Restrictions apply.

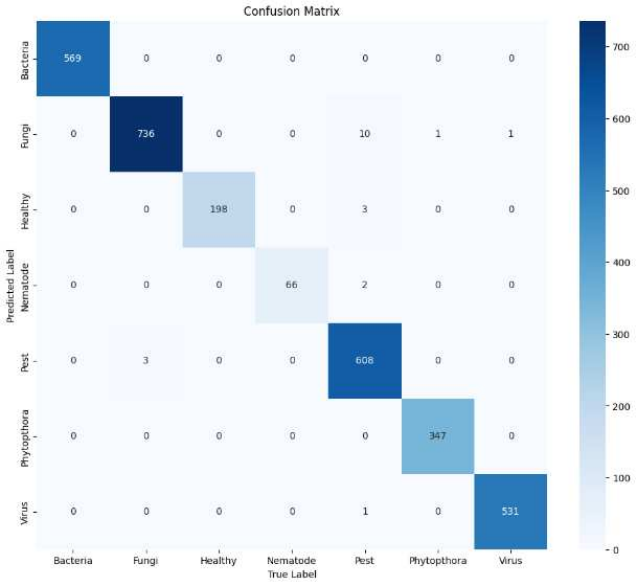

Fig. 11. ViT Model Confusion Matrix

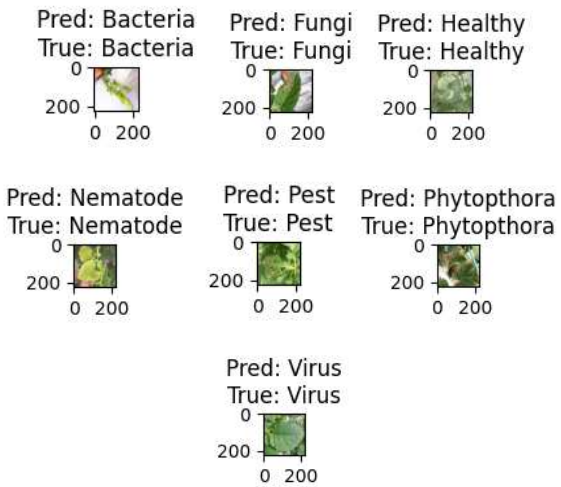

Fig. 12. Result of the ViT model.

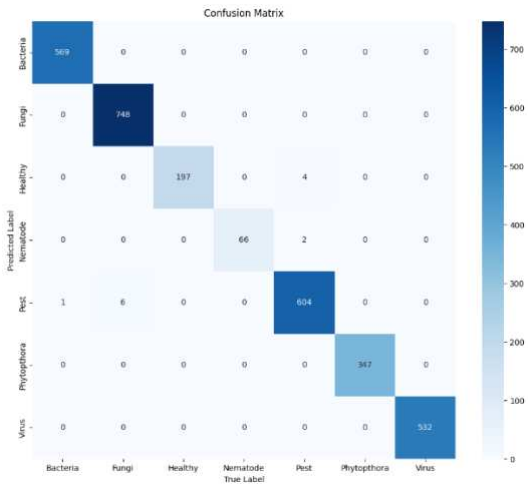

Fig. 13. CoAtNet model confusion matrix.

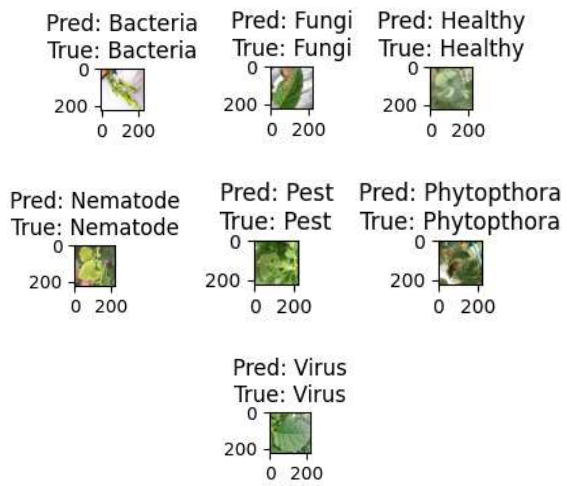

Fig. 14. Result of the CoAtNet model.

According to the confusion matrix, the CoAtNet model exceeded the performance of the ViT model in identifying potato leaf diseases. Additionally, the CoAtNet model demonstrated superior performance compared to several models from previous studies. Table IV provides a summary of the classification performance for potato leaf disease detection.

TABLE IV. SUMMARY OF SOME RELATED WORK ON POTATO LEAF DISEASE CLASSIFICATION

Work	Method	Accuracy (%)		
		Training	*Validation*	*Testing*
[10]	VGG16	-	**91.00**	**91.31**
	VGG19	-	**91.00**	**90.96**
[11]	EfficientNet	**98.04**	**95.64**	**95.92**
	ConvNeXt	**100**	**95.44**	**94.29**
[17]	CNN	-	-	**99.07**
[18]	CNN	-	**99.00**	-
Proposed	Vision Transformer (ViT)	**99.71**	**99.12**	**99.32**
	CoAtNet	**99.65**	**99.48**	**99.58**

V. CONCLUSION

This paper compared Vision Transformer (ViT) and Convolutional Attention Network (CoAtNet) for potato leaf disease classification. Both models performed exceptionally well, with ViT achieving 99.12% for validation accuracy and 99.32% for test accuracy. Also, CoAtNet slightly outperformed it at 99.48% for validation accuracy and 99.58% for test accuracy. CoAtNet's superior accuracy highlights its effectiveness in identifying potato leaf diseases with minimal error. These results demonstrate the potential of machine learning in agriculture, particularly for early disease detection. This research emphasizes the importance of developing advanced machine learning models to enhance crop monitoring, enabling timely interventions and ultimately improving agricultural productivity and food security.

ACKNOWLEDGMENT

The authors would like to thank the Laboratory of Artificial Intelligence, Control, and Automation Laboratory, Faculty of Industrial Technology, Institut Teknologi Bandung for their support during this research.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:10 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] FAO, The State of Food and Agriculture 2021, Rome, 2021.

[2] J. Bamberg and G. Greenway, “Nutritional and Economic Prospects for Expanded Potato Outlets,” American Journal of Potato Research, vol. 96, pp. 206-215, 2019. doi: https://doi.org/10.1007/s12230-018-09698-0

[3] International Potato Center, “Potato Facts and Figures.” [Online]. Available: https://cipotato.org/potato/potato-facts-and-figures/. [Accessed: June 18, 2024].

[4] H. Campos and O. Ortiz, The Potato Crop: Its Agricultural, Nutritional and Social Contribution to Humankind. Springer Nature, 2020. doi: 10.1007/978-3-030-28683-5.

[5] J. E. Munyaneza, Bizimungu and B., “Management of potato pests and diseases in Africa,” Insect Pests of Potato, pp. 407-426, 2022. doi: 10.1016/B978-0-12-821237-0.00016-0.

[6] S. Weizheng, W. Yachun, C. Zhanliang and W. Hongda, "Grading Method of Leaf Spot Disease Based on Image Processing," 2008 International Conference on Computer Science and Software Engineering, Wuhan, China, 2008, pp. 491-494, doi: 10.1109/CSSE.2008.1649.

[7] A. Dosovitskiy, “An image is worth 16x16 words: Transformers for image recognition at scale,” arXiv preprint arXiv:2010.11929, 2020. doi: 10.48550/arXiv.2010.11929.

[8] Z. Dai, H. Liu, Q. V. Le, and M. Tan, “Coatnet: Marrying convolution and attention for all data sizes,” Advances in Neural Information Processing Systems, vol. 34, pp. 3965–3977, 2021.

[9] N. H. Shabrina, S. Indarti, R. Maharani, D. A. Kristiyanti, and N. Prastomo, “A novel dataset of potato leaf disease in uncontrolled environment,” Data in Brief, vol. 52, 2024, Art. no. 109955.

[10] R. A. Sholihati, I. A. Sulistijono, A. Risnumawan and E. Kusumawati, "Potato Leaf Disease Classification Using Deep Learning Approach," 2020 International Electronics Symposium (IES), Surabaya, Indonesia, 2020, pp. 392-397, doi: 10.1109/IES50839.2020.9231784.

[11] Irmawati, D. A. Kristiyanti, T. A. M, N. H. Shabrina, S. Indarti and N. Prastomo, "Early Detection of Potato Leaf Pest and Disease Using EfficientNet and ConvNeXt Architecture," 2023 7th International Conference on New Media Studies (CONMEDIA), Bali, Indonesia, 2023, pp. 167-172, doi: 10.1109/CONMEDIA60526.2023.10428527.

[12] M. Islam, Anh Dinh, K. Wahid and P. Bhowmik, "Detection of potato diseases using image segmentation and multiclass support vector machine," 2017 IEEE 30th Canadian Conference on Electrical and Computer Engineering (CCECE), Windsor, ON, Canada, 2017, pp. 1-4, doi: 10.1109/CCECE.2017.7946594.

[13] M. S. Hylmi, Wiharto and E. Suryani, "Detection of Potato Leaf Disease Using Multi-Class Support Vector Machine Based on Texture, Color, and Shape Features," 2022 International Conference on Electrical and Information Technology (IEIT), Malang, Indonesia, 2022, pp. 20-24, doi: 10.1109/IEIT56384.2022.9967866.

[14] Y. P. Wasalwar, K. S. Bagga, V. K. Joshi and A. Joshi, "Potato Leaf Disease Classification using Convolutional Neural Networks," 2023 11th International Conference on Emerging Trends in Engineering & Technology - Signal and Information Processing (ICETET - SIP), Nagpur, India, 2023, pp. 1-5, doi: 10.1109/ICETET-SIP58143.2023.10151591.

[15] A. Srivastava, B. S. Rawat, P. Bajpai and S. A. Dhondiyal, "Potato Leaf Disease Detection Method Based on the YOLO Model," 2024 4th International Conference on Data Engineering and Communication Systems (ICDECS), Bangalore, India, 2024, pp. 1-5, doi: 10.1109/ICDECS59733.2023.10502511.

[16] A. Bajpai, M. Tyagi, M. Khare and A. Singh, "A Robust and Accurate Potato Leaf Disease Detection System Using Modified AlexNet Model," 2023 9th International Conference on Computer and Communication Engineering (ICCCE), Kuala Lumpur, Malaysia, 2023, pp. 264-269, doi: 10.1109/ICCCE58854.2023.10246064.

[17] S. K. Hoysala, S. K. S., S. D. R., S. S. Chandra and A. P. B. V., “Recognition and Classification of Leaf Disease in Potato Plants,” Journal of Data Mining and Management, vol. 4, no. 2, pp. 25-29, 2019. doi: : 10.5281/zenodo.3353806

[18] S. Bangari, P. Rachana, N. Gupta, P. S. Sudi and K. K. Baniya, "A Survey on Disease Detection of a potato Leaf Using CNN," 2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS), Coimbatore, India, 2022, pp. 144-149, doi: 10.1109/ICAIS53314.2022.9742963.

[19] X. Liu and C. Aldrich, “Flotation Froth Image Recognition Using Vision Transformers,” IFAC-PapersOnLine, vol. 56, no. 2, pp. 2329–2334, 2023. doi: 10.1016/j.ifacol.2023.10.1202.

[20] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov and L. -C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, Salt Lake City, UT, USA, 2018, pp. 4510-4520, doi: 10.1109/CVPR.2018.00474.

[21] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” Advances in Neural Information Processing Systems, vol. 30, 2017. doi: 10.48550/arXiv.1706.03762.

[22] Google, “Hugging Face,” [Online]. Available: https://huggingface.co/google/vit-base-patch16-224. [Accessed 18 June 2024].

[23] R. Wightman, “Hugging Face,” [Online]. Available: https://huggingface.co/timm/coatnet_bn_0_rw_224.sw_in1k. [Accessed 19 June 2024].

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:10 UTC from IEEE Xplore. Restrictions apply.